Final outline proposal: 15-minute slide deck

EU ETS and CBAM, the Germany steel case. Working title: "How EU ETS and CBAM Can Strengthen Europe's Long-Term Green Manufacturing Economy: The Case of Germany". On-slide subtitle: "Germany, iron and steel; descriptive evidence."

Status: final proposal for the team, 22 October 2026. Built from the LINE chat and the four Word files, then updated with live data pulls, status checks and German primary sources made on 22 October 2026, and checked by three independent reviews (numbers, consistency, red team). Key numbers carry a source tag; the tag list is in backup page A15. Nothing here is an estimate of an ETS or CBAM effect.

1. Decisions and what changed

Item	Decision	Note
Slot	15 minutes of speaking, Q&A after.	The course brief says 20 + 10; a 20-minute stretch plan is in section 8.
Retitled scope	Approved by the requester: ETS + CBAM, Germany, iron and steel. The outline and the retitle were confirmed by the requester on 26 Oct 2026, and the slide sketches (`outline/sketches/`) were started.	The chat shows lecturer approval only for the original CBAM-only topic (relayed on 7 Oct); no lecturer approval of the retitle or of the 15-minute plan is on record. The topic was cleared remotely in October, so send both now through the same channel and record the answer.
Theory diagrams	Fig. 1 (unchanged) and the redrawn Fig. 2 approved for the deck.	Only the lecturer's check of hand-drawn graphs needs the on-site class. Each diagram slide keeps a text fallback with the same timing.

Item	Decision	Note
Language	Slides in English; spoken language follows the lecturer.	Assumption: sources, lecture PDFs and the assignment brief are English.
Phase-out of free allocation in CBAM sectors	Position recorded 22 Oct 2026: support slowing the phase-out as the Commission proposed on 17 Jul 2026 (15% still granted in 2034–37, end 2038), with the 2031 investment-plan link kept.	Team judgement, shown on S13; the proposal is not law and the Council has not decided. Recorded from the requester's instruction; the LINE chat holds no discussion of it, so the other members confirm it before the lecturer is told "the group supports".
Template	16:9, white background, **TH Sarabun New** for all text (slide body at least 28 pt), blue = before, green = change, amber = caution, grey = footers.	Member D builds the slides and may substitute a department template. Fig. 1 is the workbook's own image and keeps its original font.

Corrections to the research workbook found in this pass

These change what the deck may say; each is already applied below.

1. **Electricity (slide 5, backup A10).** The workbook's 5.92 ct/kWh for 2020 is the SMARD price with all possible reductions, not without. The no-reductions price was about 17.1 ct/kWh in 2020, so "nearly triple the 2020 level" is wrong: the no-reductions price in 2024 (16.1 in the 22 Oct 2026 SMARD vintage, 16.8 in the Feb 2025 vintage) is slightly below 2020. The "+77%" figure is a with-reductions comparison and exists only in the Feb 2025 vintage (+65% in the current vintage). Invoiced prices for the largest consumers (150,000 MWh a year and more) did rise, from 6.88 to 13.01 ct/kWh in 2020–24 (+89%).
2. **Direct versus by-origin imports (slide 7).** Direct extra-EU Chapter 72 imports (cleared at German customs) were only 37% of German non-EU-origin imports in 2025 (66% in 2019), mostly a change since 2022 whose cause is not identified. The direct series

fell 56% (2019–25); non-EU steel counted by origin, including what arrives via other EU states, fell 22%.

3. **Steel trade after 2024 (slide 8).** WV Stahl's 2026 edition revises 2024 and adds 2025: the all-partner balance narrowed from +4.2 Mt (2024) to +1.9 Mt (2025) as imports rose (+7.9%) and exports fell (–3.9%). Since 2017 imports are still down more than exports (–28% against –18%).
4. **Public support (slides 9, 11, backup A11, A13).** Salzgitter's aid is about EUR 1.3 bn since February 2026 (EUR 1.076 bn paid by 30 Jun 2026), not EUR 1 bn; Thyssenkrupp's "full hydrogen by 2029" was a 2023/2024 company target and is stale; the carbon-contract thresholds are now 50% within four years and 85% at the end, not 60% / 90%; EUR 6.9 bn was a July 2024 commitment total (about EUR 5.9 bn live); EUR 12.5 bn total cost and 11.5 Mt a year could not be verified.

Open items from the first proposal and their results

Open item	Result	Where it lands
Redraw Fig. 2	Done: partial pass-through, no invented numbers; the supply shift is labelled "size not evidenced".	S5
Extra-EU steel exports	Pulled for 1999–2025 (Comext, vintage 16 Oct 2026); Destatis by destination agrees within 0.1% in 2017–25.	S8, S10, A6
Materiality	Direct extra-EU imports are 8.1% of German Chapter 72 import tonnes in 2024 (6.1% in 2025); non-EU origin 17.3% in 2024.	S7
Steel series (outcome)	Electric-arc share 27.7–32% since 2020, no trend; CO2 per tonne shows no sustained fall (ETS-derived –2.5% since 2019, WV Stahl +1.5%).	S10, S12, A8
Apparent steel consumption	WV Stahl publishes market supply of rolled steel: 40.8 Mt (2017) to 29.1 Mt (2025).	S8, A6
Domestic steel price	Destatis producer price index for pig iron and steel pulled.	S10

Open item	Result	Where it lands
Tax	ETS auction receipts (all sectors, EUR 5.4 bn in 2025) go to the climate fund; CBAM sales start 1 Feb 2027; corporate and payroll effects are not measurable.	S10, S13, A6
Certificate price	Q1–Q3 2026 confirmed. **No Q4 price can exist before 4 Jan 2027**, after the talk.	S4, S5, A3
Stegra delivery status	Checked: more capital needed, no published start date, no confirmed delivery to Mercedes-Benz or BMW.	S9, S11, A13
Downstream extension	Still a proposal; Parliament position 15 Sep 2026; negotiations under way.	S13
Steel quota text	Resolved: 18.3 Mt, 50% out-of-quota duty, about 47% below 2024 quotas; Reg. (EU) 2026/1384, applies 1 Jul 2026.	S4, A5
2026 trade data	Jan–Aug 2026 available (provisional), compared only with Jan–Aug 2025.	S5, S7, S8
Provisional GVA flag	Rechecked: still provisional for 2023–2025; no value changed.	S1, S6
German primary sources	WV Stahl, Destatis, SMARD and German support amounts sourced (section 6).	S5, S7–S11
Lecturer confirmation (retitle, slot, diagrams)	Recorded as approved by the requester; send the retitle and slot through the topic channel; the hand-drawn graphs need the on-site class.	Section 8

2. The talk in one view

Block	Slides	Minutes	Checkpoint
CH1 Policy details	S1–S4	4.0	4.0
CH2 Economic effects	S5–S11	8.5	12.5
CH3 Conclusion and suggestion	S12–S13	2.5	15.0
Backup (not timed)	A1–A15	—	Q&A

#	Min	Slide	Message in one line
S1	1.0	Germany, iron and steel: what does the record show since EU ETS and CBAM?	The question, scope and method; why Germany and why steel.

#	Min	Slide	Message in one line
S2	1.0	Why price carbon	Emissions are a cost society bears (Fig. 1); before / during / latest facts.
S3	1.0	Two tools, one logic	ETS inside the EU, CBAM at the border; purpose: prevent carbon leakage.
S4	1.0	Timeline	2005 to 2034; CBAM payments start in 2027; the steel quota is a separate measure.
S5	1.0	Theory: a carbon cost on imports	Partial pass-through (Fig. 2); before / during / after facts; no after-effect claim.
S6	1.0	German manufacturing, 1995–2025	Output grew through the ETS launch; a graph cannot separate policy from other shocks.
S7	1.5	Steel imports from outside the EU	Direct imports −56% (2019–25), by origin −22%; overlapping events.
S8	1.5	Steel trade balance	Two universes; the all-partner balance narrowed in 2025; no verdict.
S9	1.0	What firms signed	Plans, not delivered tonnes; public aid cannot be separated from policy anticipation.
S10	1.5	Checklist: all eleven variables	One line each; context indicators or explicit gaps; no effect identifiable.
S11	1.0	Costs and conditions	A premium or gap remains; benefits are conditional.
S12	1.0	Conclusion	The record is consistent with a move toward cleaner steel; causation and per-tonne gains not shown.
S13	1.5	Suggestion	Keep the framework, change the delivery; named triggers to reconsider.

Rules on every slide: one dominant evidence label (Observed data / Policy rule / Model-based result / Theoretical-illustrative / Company announcement / Team judgement); row-level chips may

also say "Context only" or "Gap"; at most one chart and one small table or tile (a two-panel figure counts as one chart); table cells of about eight words, with detail moved to speaker notes and backup; no title says "effect"; policy dates are annotations, not proof; a CH1 | CH2 | CH3 ribbon; the 11 required variables (X, M, NX, BOP, Q, P, GDP, tax, C, I, G) are lit on the slide that shows them, and BOP, GDP, tax, C and G are first lit on S10.

3. CH1 — Policy details (4.0 min)

S1 · 1.0 min · Germany, iron and steel: what does the record show since EU ETS and CBAM?

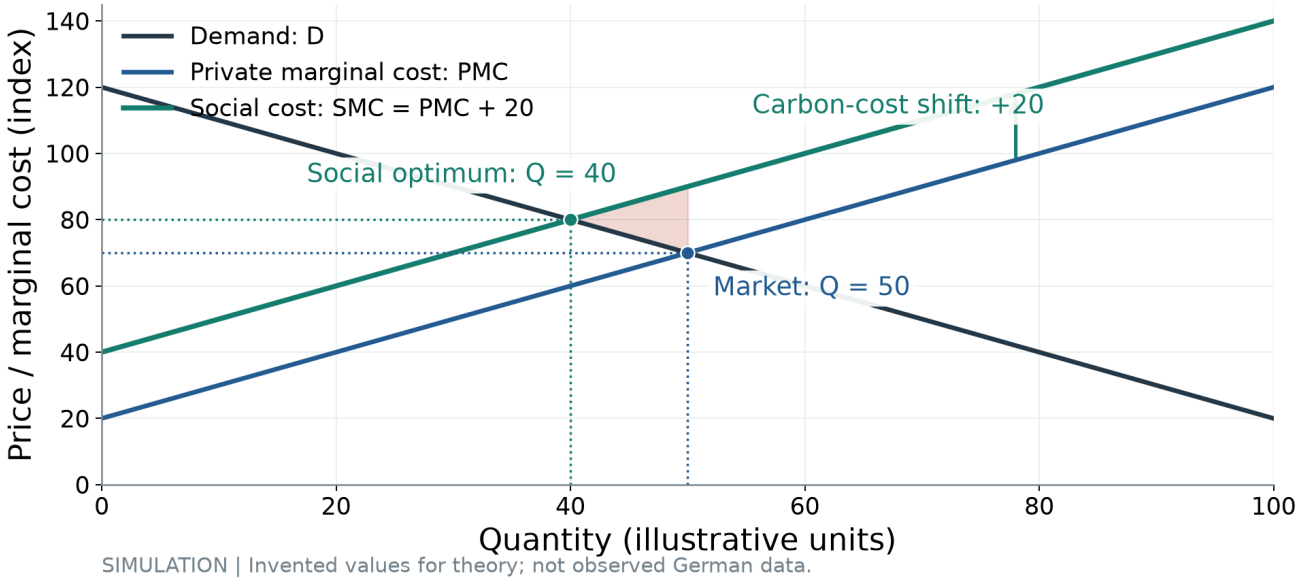
- **Message:** We ask what the record shows for German steel since EU ETS and CBAM, and which opportunities and adjustment costs appear; descriptive, no causal claim.
- **On the slide:** question and roadmap; three tiles with definitions printed. Manufacturing is 19.9% of German value added in 2024 (provisional) against 15.9% for the EU27; 19.4% in 2025 (provisional) [U22]. Germany holds 26% of EU sold industrial production, 2025, a production share and not GDP [U26]. German crude steel 34.1 Mt in 2025 (WV Stahl / Destatis, 2026 edition; revised to 34.3 Mt in Aug 2026) [Dataset A].
- **Say:** steel is the first-stage input to automotive and machinery and a CBAM-covered sector [S4]; CBAM is EU law and Germany applies it [S1][S2]; method is graphs, course theory and dated company cases, with no regression.
- **Limits:** the EU27 figure for 2025 is not published yet (Bulgaria and Latvia missing), so the comparison stays at 2024.
- **Label:** Observed data.

S2 · 1.0 min · Why price carbon: emissions are a cost society bears (lecturer check pending)

- **Message:** Steelmaking emissions cost society more than the producer pays; a carbon price makes producers face that cost at the margin, which is the logic of the ETS.
- **On the slide:** Fig. 1 below, unchanged, with its SIMULATION footer. Beside it a before / during / latest strip. BEFORE:

manufacturing production index rose from 71.3 to 85.7 over 1995–2004 (2021 = 100) [U23]. DURING: 2005 launch; the pilot 2005–07 used mostly free allocation and was not a uniform charge [U1]. LATEST: covered EU power and industry emissions were about 50% below 2005 in 2024, an EU aggregate led by the power sector; not German steel and not CBAM [U29].

- **Say:** the price is a marginal incentive; free allocation covered about 94% of German iron-and-steel emissions in 2024 (adjusted for process gases passed on), so the net bill was small and CBAM's phase-in is what changes the border comparison [DEHSt VET 2024].
- **Fallback:** a two-line wedge statement plus the same strip.
- **Label:** Theoretical-illustrative.



S3 · 1.0 min · Two tools, one logic: ETS inside, CBAM at the border

- **Message:** ETS prices emissions inside the EU; CBAM applies a comparable carbon-cost logic to covered imports by embedded emissions; it is neither a ban nor a flat tariff.
- **On the slide:** five-row table, cells of about eight words. What: declining cap, tradable allowances | carbon cost on covered imports by embedded emissions. **Purpose:** cap and price emissions | prevent carbon leakage (production or emissions shifting abroad). Who surrenders: installations surrender allowances | authorised declarant surrenders certificates. Price:

auction price | average of EU ETS auction prices, quarterly in 2026, weekly from 2027. Start: 2005 pilot, mostly free allocation | reporting Oct 2023, definitive 1 Jan 2026.

- **Say:** free allocation lowers net cost but is not an exemption [U1] [U2]; German auction prices averaged EUR 14.9 a tonne in 2018, 83.7 in 2023 and 73.9 in 2025 [UBA / EEX]; Germany sets no tariff of its own [S1][S2]. Coverage and the 50-tonne threshold are in A4; the worked example is in A3.
- **Label:** Policy rule.

S4 · 1.0 min · Timeline: ETS since 2005, CBAM payments from 2027

- **Message:** Carbon pricing arrived in stages and CBAM's cash payments start in 2027; our annual data end in 2025 and monthly trade data run to August 2026, so no data point includes a paid CBAM charge.
- **On the slide:** two-lane timeline. Lane 1 climate policy: 2005 ETS launch [U1]; 14 Jul 2021 CBAM proposed [S3]; 4 Oct 2022 and 20 Jul 2023 EU approval of German aid for Salzgitter and Thyssenkrupp; 1 Oct 2023 reporting [U3][S7]; 1 Jan 2026 definitive regime [S6]; 1 Feb 2027 certificate sales and 30 Sep 2027 first surrender [U35][S5] (the Q4 2026 price is due 4 Jan 2027); 2034 end of the phase-in under current law. Small step chart of the CBAM factor, 97.5% (2026) to 0% (2034) [S9][S10], with a footnote: a Commission proposal of 17 Jul 2026 would keep 15% for 2034–37 and end at 2038; not law, Council debated it on 12 Oct without a decision; our suggestion on it is on S13. Lane 2, a separate trade measure that is not CBAM: Regulation (EU) 2026/1384, Council adoption 8 Jun 2026, applies 1 Jul 2026; duty-free quota 18.3 Mt a year, about 47% below the 2024 safeguard quotas; out-of-quota duty 50% (the safeguard's was 25%).
- **Speaker note:** data windows are manufacturing 1995–2025; extra-EU Chapter 72 trade 1999–2025 annual plus Jan–Aug 2026 monthly (provisional); steel trade all partners 2017–2025; prices to September 2026.
- **Label:** Policy rule.

4. CH2 — Economic effects (8.5 min)

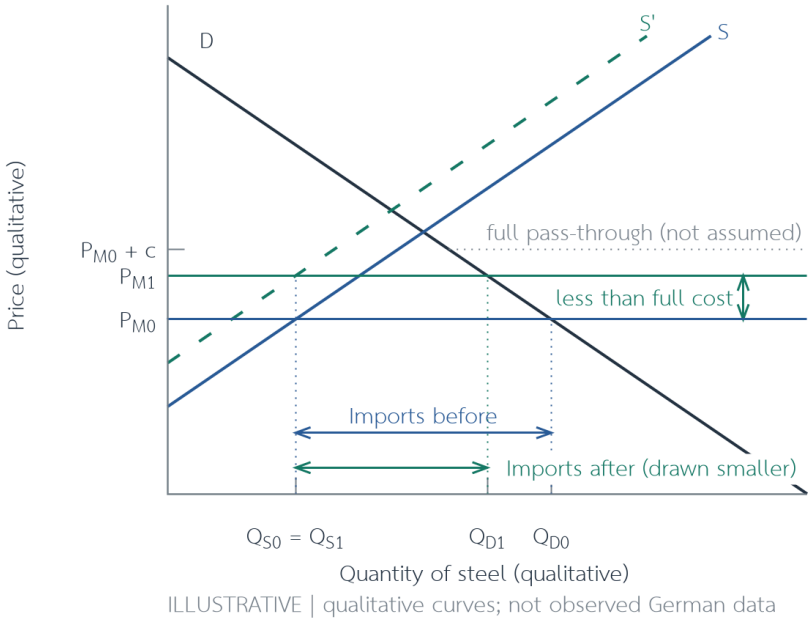
S5 · 1.0 min · Theory: a carbon cost on imports, but not one-for-one (lecturer check pending)

- **Message:** A carbon-related import cost narrows the import price advantage, but pass-through need not be full and domestic costs may also change, so output and imports respond in an open, short-run way.
- **On the slide:** the redrawn Fig. 2 below (qualitative, no numbers, ILLUSTRATIVE banner). Beneath it a before / during / after strip with real facts, imports shown both ways (direct = cleared at German customs; by origin = all channels):
- BEFORE: direct imports 4,624 kt (2017) to 2,697 kt (2022); by origin 6,933 to 4,704 kt [Comext, Destatis]. Modelled industrial electricity price without reductions 17.1 ct/kWh (2020) to 35.9 (2022) [SMARD, vintage of 22 Oct 2026].
- DURING (reporting from 1 Oct 2023; 2023 straddles the start and is not assigned): direct imports 1,955 kt (2024) to 1,464 kt (2025); by origin 4,156 to 3,919 kt. Electricity 16.1 ct/kWh in 2024 and 16.5 in 2025.
- AFTER (2026, provisional): Jan–Aug direct imports 1,108 kt against 1,028 kt in Jan–Aug 2025, +7.7% in tonnes and −0.7% in value [Comext]; by origin Jan–Jul +6.1% [Destatis]; electricity 17.3 ct/kWh for Jan–Sep 2026; CBAM certificate prices EUR 75.36, 75.28 and 82.32 per tCO2e for Q1–Q3 2026, Q4 due 4 Jan 2027 [Dataset E]; first payments 2027; no after-effect claim.
- **Say up front (answers the CH2 brief):** because no CBAM payment exists in our data, CH2 gives theory predictions, descriptive context and a watch-list from February 2027, not an estimated effect. "Our talk falls inside Q4, so the Q4 price does not exist yet." By origin, imports were flat inside the reporting window (3,799 kt in 2023, 3,919 in 2025), so part of the direct fall reflects where steel is cleared.
- **Limits:** SMARD is a Bundesnetzagentur model, not invoiced prices; it states no consumption band and is not steel-specific. Invoiced prices for the largest consumers rose 89% in 2020–24, so energy costs did rise for big users; never say "steelmakers

paid" or "tripled". Net ETS cost is small under free allocation, so the supply shift in Fig. 2 is a possibility, not a finding.

- **Fallback:** a mechanism box with the same strip.
- **Label:** Theoretical-illustrative; the electricity strip is a Model-based result. **Chips lit:** M, Q, P.

Redrawn Fig. 2: carbon-related import cost, partial pass-through



What moves

Landed import price: P_{M0} to P_{M1}, up by less than the full carbon cost c

Domestic supply may shift to S' if ETS, energy and capital costs rise (size not evidenced)

Quantity demanded falls along D (Q_{D0} to Q_{D1})

Domestic output: may rise, stay or fall (drawn unchanged)

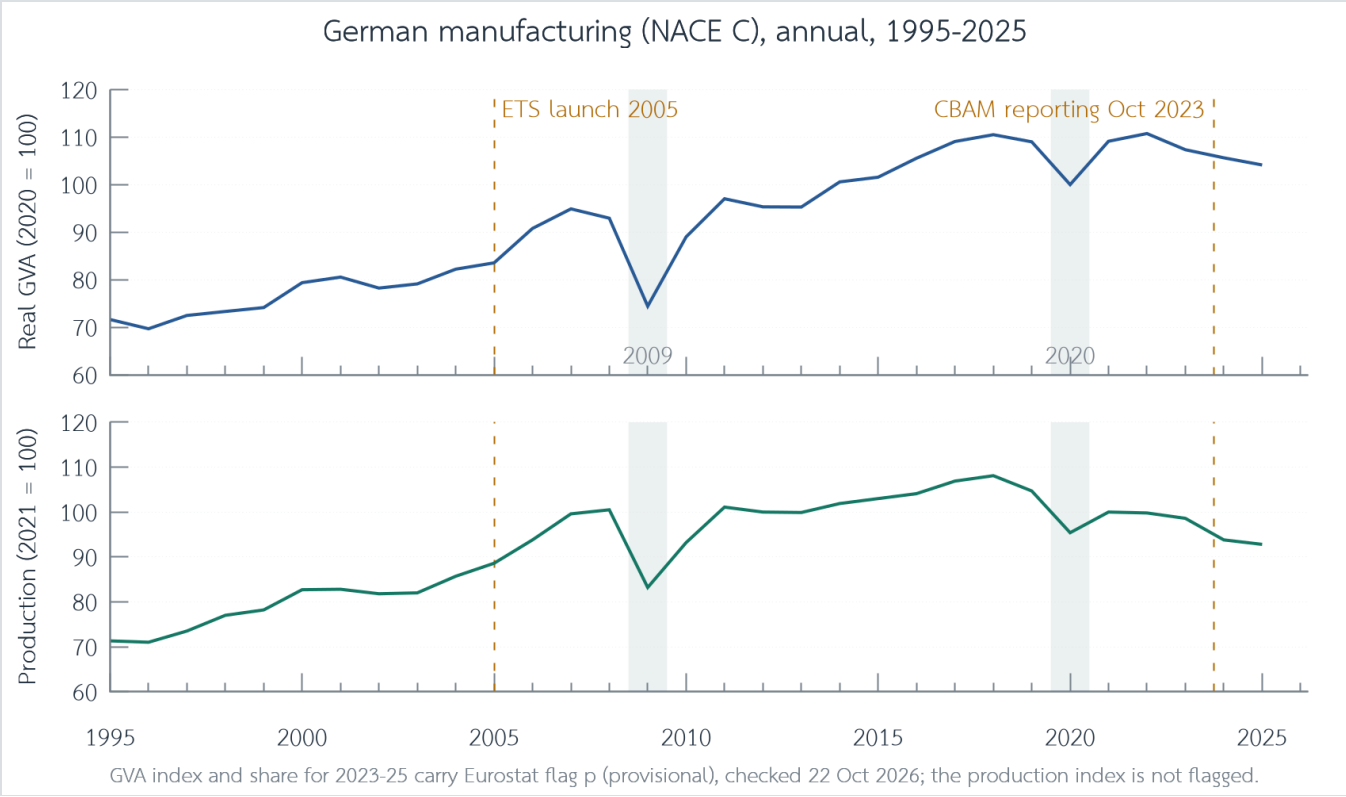
Import gap: drawn smaller, not guaranteed

Short run, static, one point in time. Steel-using manufacturers face the cost rise downstream (not drawn).

ILLUSTRATIVE | qualitative curves; not observed German data

S6 · 1.0 min · German manufacturing, 1995–2025

- **Message:** Real manufacturing output grew through the ETS launch, fell sharply in 2009 and 2020 and has softened since 2022; a graph alone cannot separate policy from energy, demand and COVID.
- **On the slide:** the chart below; tile: manufacturing share of total GVA 22.3% (1995) to 19.4% (2025), EU27 15.9% (2024), value added and not GDP. Footnote: GVA index and share for 2023–25 are provisional (Eurostat flag p, checked 22 Oct 2026); the production index is not flagged. Since the Destatis revision of 25 Aug 2026 real manufacturing GVA is extrapolated with a deflated turnover index, so the two panels are different indicators and can diverge. Primary source: Destatis (the Eurostat series is Destatis data; all 31 years match).
- **Footer:** does not show an ETS or CBAM effect, or no effect.
- **Label:** Observed data. **Chip lit:** Q only. **Tags:** U22, U23.

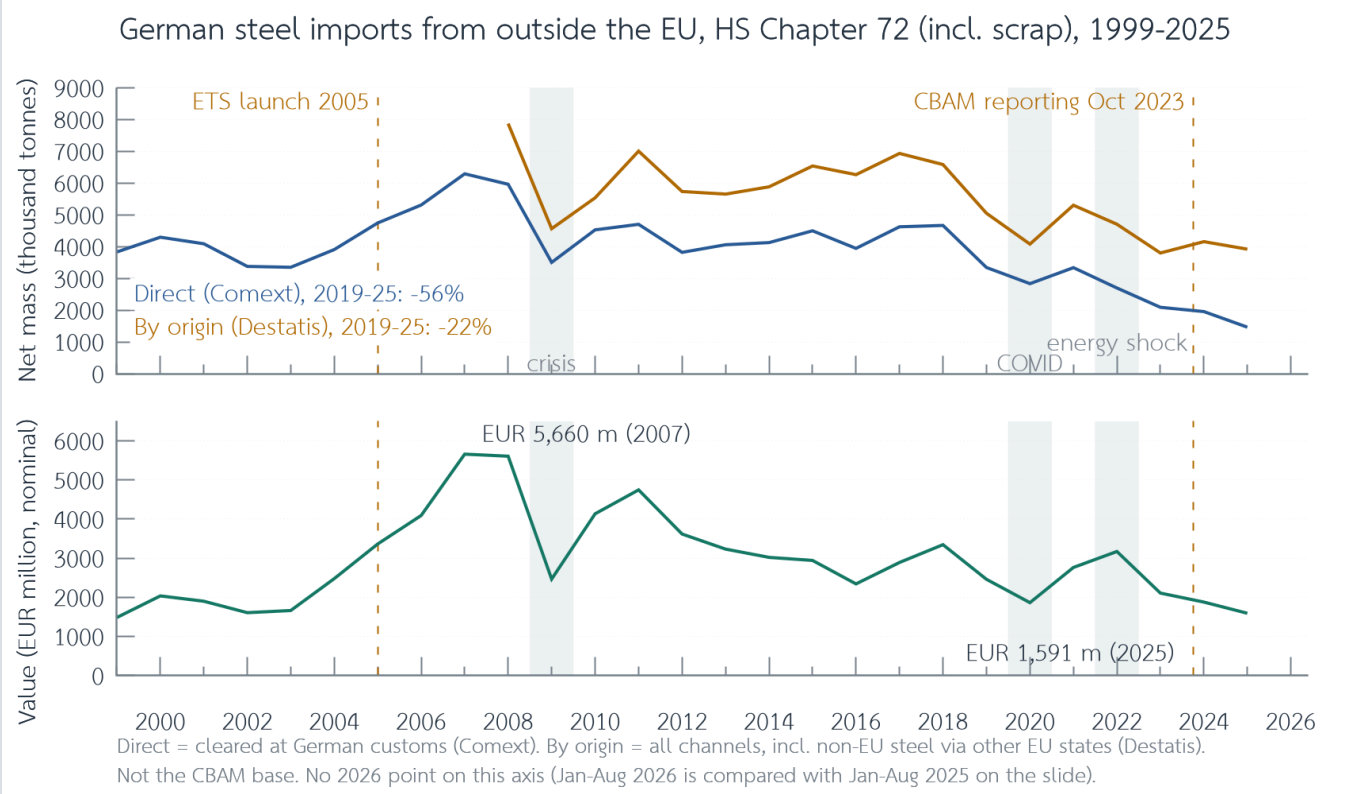


S7 · 1.5 min · Iron and steel (HS Chapter 72, incl. scrap) imports from outside the EU

- **Message:** Direct extra-EU imports rose to 6,291 kt in 2007, were 3,342 kt in 2019 and fell to 1,464 kt in 2025 (–56%); counted by origin, including steel that arrives via other EU states, the fall is –22% (5,050 to 3,919 kt). Overlapping events prevent reading either as a policy effect.
- **On the slide:** the chart below, both lines on the tonnes panel and direct value on the lower panel. Caption: unit value EUR/t 736 (2019), 1,176 (2022), 1,087 (2025), a product-mix indicator and not a price index [Dataset C]. Small inset, labelled provisional and same months: Jan–Aug 2025 against Jan–Aug 2026, direct imports 1,028 to 1,108 kt (+7.7%), EUR 1,127 m to 1,119 m (–0.7%).
- **Say:** imports rose after the 2005 launch to the 2007 peak, so we do not claim ETS cut imports. Materiality: direct imports are 8.1% of German Chapter 72 import tonnes in 2024 (6.1% in 2025, 13.4% in 2017); by origin, non-EU steel is 17.3% in 2024, and that share has been flat at 16–18% since 2019, so there is no visible shift of sourcing to EU suppliers. The direct series was 66% of the by-origin series in 2019 and 37% in 2025, mostly a change since

2022 whose cause is not identified. Scrap is 27% of 2025 direct imports. Heading 7203 (direct-reduced iron) fell from 1,016 kt in 2017 to 28 kt in 2025 directly, but by origin only from 1,069 to 522 kt (–51%), so much of the direct fall is routing. Do not quote single months (year-on-year swings run from –44% to +49%).

- **Limits:** Chapter 72 includes scrap, pig iron and DRI and omits CBAM-covered Chapter 73 goods and ore. Not the CBAM base. The by-origin series starts in 2008. 2023 was revised slightly in the 16 Oct 2026 vintage (rounds the same).
- **Label:** Observed data. **Chips lit:** M, Q, P. **Tags:** Comext DS-045409, Destatis 51000-0007, U24, U25, Dataset C.

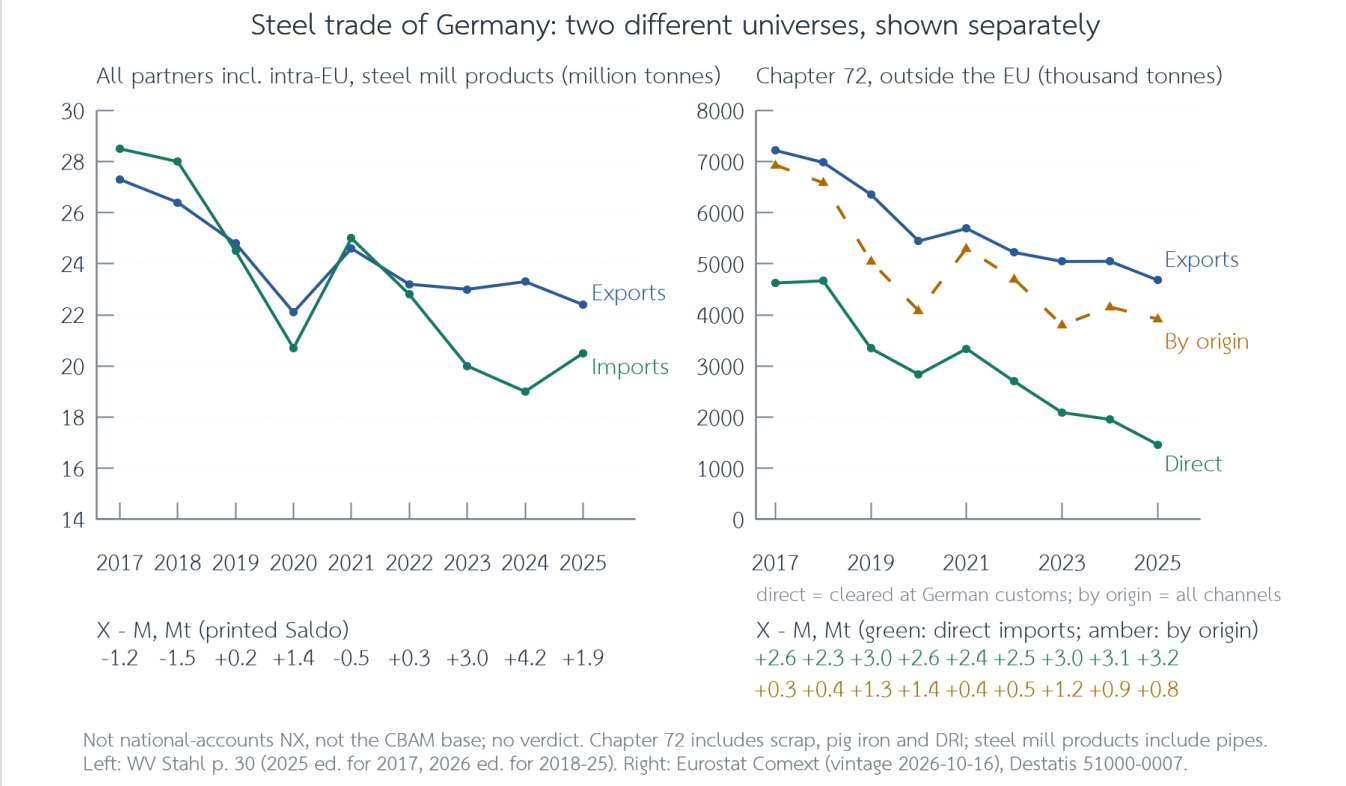


S8 · 1.5 min · Steel trade balance: what the matched series show

- **Message:** Through 2024 steel imports fell faster than exports and the German balance rose; in 2025 imports rose and the all-partner balance narrowed to +1.9 Mt. Outside the EU the balance is +0.8 Mt on the like-for-like by-origin concept; the +3.2 Mt with direct imports only reflects steel cleared in other EU states. Different universes; no competitiveness verdict.
- **On the slide:** the two-panel chart below (never subtracted from each other). Left: all partners incl. intra-EU, steel mill products,

2017–2025, exports 27.3 to 22.4 Mt and imports 28.5 to 20.5 Mt, printed balance −1.2, −1.5, +0.2, +1.4, −0.5, +0.3, +3.0, +4.2, +1.9 Mt [WV Stahl p. 30, 2025 edition for 2017 and 2026 edition for 2018–25]. Right: Chapter 72 to and from outside the EU, 2017–2025: exports 7,217 to 4,681 kt [Comext; Destatis agrees within 0.1% in 2017–25], imports direct 4,624 to 1,464 kt, imports by origin 6,933 to 3,919 kt. One-line tile, provisional: Jan–Aug 2026 extra-EU exports −16%, direct imports +8%; all-partner Chapter 72 exports +5% and imports +2% [Comext, Destatis]; causes not tested.

- **Say:** Member B's point: imports alone say nothing about competitiveness, so we add exports. NX is X minus M; here it is steel only, in tonnes. Extra-EU exports fell 35% from 2017 to 2025 and are the lowest in 27 years; EU-27 exports of steel mill products to third countries fell 41% (different product list), so the fall is not German-specific. Behind the trade, German market supply of rolled steel fell from 40.8 Mt (2017) to 29.1 Mt (2025) [WV Stahl]; the 2025 uptick rested only on stock building. Goods trade is not the balance of payments.
- **Limits:** Chapter 72 includes scrap, pig iron and DRI; steel mill products include pipes (probably mostly HS 73; no HS list is published) and are 76–85% of Chapter 72 tonnes; levels differ. Partner scope of the WV Stahl table is inferred (all partners incl. intra-EU). Not national-accounts NX (goods and services, EUR). Not the CBAM base. State the import concept (direct or by origin) every time a balance is quoted.
- **Label:** Observed data. **Chips lit:** X, M, NX (BOP footnote). **Tags:** Comext, Destatis, WV Stahl, S18.

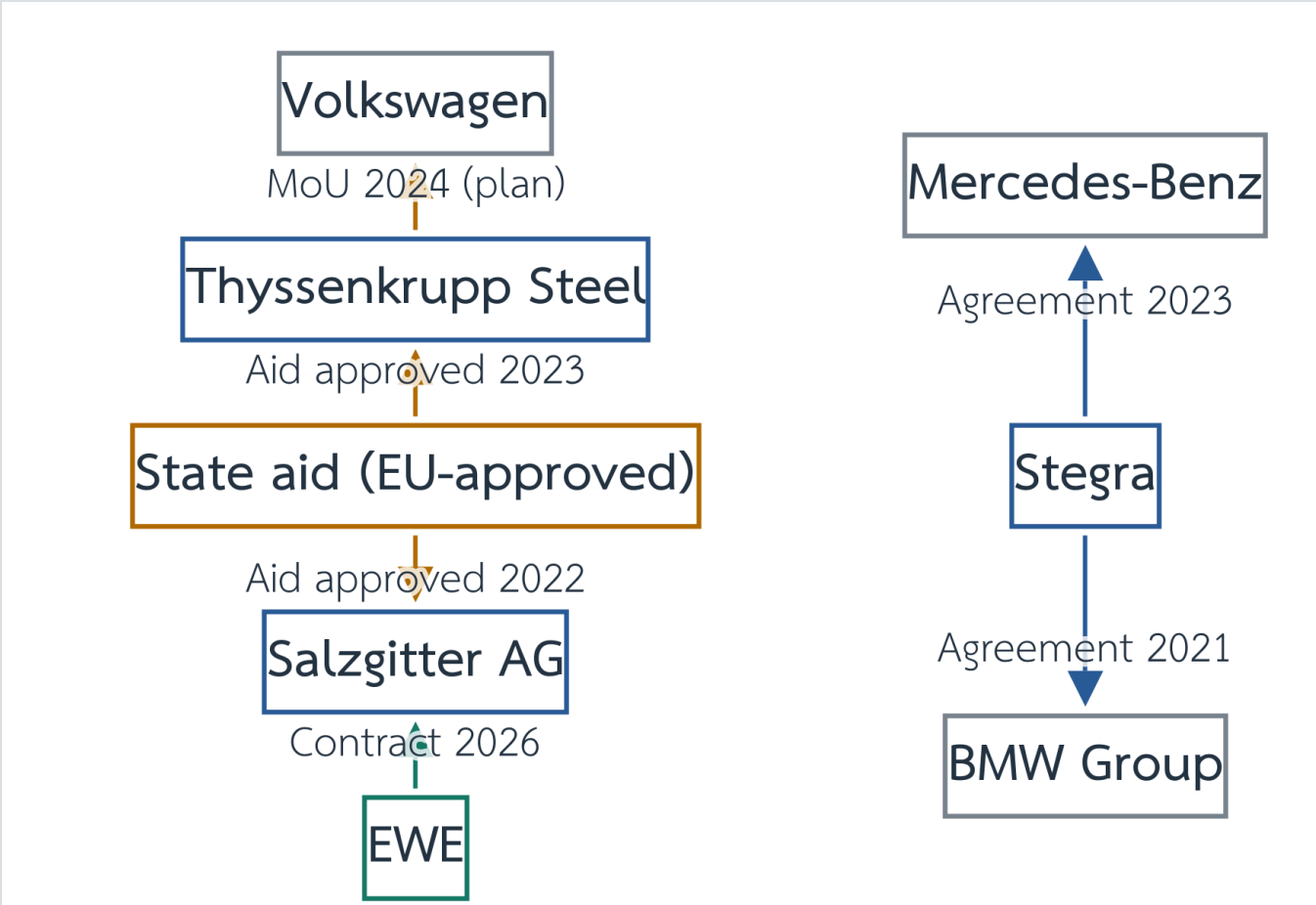


S9 · 1.0 min · What firms signed: plans, not yet delivered tonnes

- **Message:** Buyer and producer agreements show demand for lower-emission steel and announced plans; they are plans and contracts, not delivered tonnes, cost parity or a CBAM effect.
- **On the slide:** three case rows, each with a status chip, one green Opportunity line and one amber Limit line. 1. **Thyssenkrupp Steel × Volkswagen.** Opportunity: MoU of 22 Oct 2024; direct-reduction plant commissioning from 2027, supply from 2028 [U10]. Limit: a future-supply MoU, not delivery; the plant starts on natural gas and the hydrogen date is open; bluemint is a product family, not all hydrogen steel [U10][U11]. 2. **Salzgitter SALCOS × EWE.** Opportunity: first direct-reduction and electric-arc production in 2027 (green steel offered from summer 2027), full conversion mid-2030s [U12]; EWE green hydrogen contract of 9 Jun 2026, 10,000 t a year for seven years [U13]. Limit: stage 1 slipped from end-2025 to 2027; EWE covers about 6.5% of SALCOS hydrogen need; Salzgitter says new energy inputs currently cost more than coal [U12][U13]. 3. **Stegra (producer) with Mercedes-Benz (buyer, Jun 2023, about 50,000 t a year) and BMW (buyer, announced 20 Oct 2021, final contract Aug**

2022) [U14][U15]. Limit: Stegra says more capital is needed to finish Boden; the 2025 start dates have passed, no new start date is published and no delivery to either buyer is confirmed [U16].

- **Footer:** two of the four agreements pre-date reporting; direction, not cause. Public aid cannot be separated from the policies: the approvals of EUR 1.0 bn for Salzgitter (4 Oct 2022) and about EUR 2 bn for Duisburg (20 Jul 2023) pre-date reporting but post-date the 2021 CBAM proposal and the ETS price rise; the SALCOS top-up (Feb 2026) and the payments to mid-2026 (EUR 1.08 bn paid by 30 Jun 2026) do not. ArcelorMittal withdrew its German plans in June 2025; its approved EUR 1.3 bn was not used. Does not show delivered tonnes or cost parity. Volvo–SSAB and Outokumpu are backup only (A13).
- **Speaker note, not on the slide:** Stegra's own review also says the timeline "remains as expected" and that commitments to customers remain; use "not confirmed", never "none delivered".
- **Label:** Company announcement. **Chip lit:** I.
- **Sketch (26 Oct 2026):** the case rows are condensed to one Opportunity and one Limit line each and sit beside a Gephi map of the agreements (the chart of this slide): blue producers, grey buyers, green hydrogen supplier, amber public aid, arrows for announcements and approvals, no edge a verified delivery. Nodes and edges: `graphs/gephi`; the map is drawn by `graphs/gephi/render_network.ps1` (Gephi Toolkit 0.11.3).



S10 · 1.5 min · CH2 checklist: all eleven variables, one line each

- **Message:** Each required variable has a conditional mechanism and one context indicator or an explicit gap; none is read as a policy effect, and no effect is identifiable yet.
- **On the slide:** one table, revealed in three blocks (trade; market; macro and fiscal), one figure per cell. Status chip per row: Observed / Context only / Gap.

Var	Theory expects	What we can show
X	No CBAM relief on exports; steel users face dearer inputs	Exports, all partners, 27.3 to 22.4 Mt (2017–25)
M	Costly imports lose their edge	Non-EU origin 5,050 to 3,919 kt (2019–25)
NX	X – M; a steel-only change is not total NX	Steel balance +1.9 Mt (2025), all partners
BOP	Goods trade is one part of the current account	Current account about EUR 203 bn (2025), not a steel effect

Var	Theory expects	What we can show
Q	Imports and domestic output both ambiguous	Crude steel 43.3 to 34.1 Mt; electric-arc share 28–32%
P	Landed cost up, pass-through partial	Steel producer prices (2021 = 100) 101.2 (2025), 107.9 (Sep 2026)
GDP	An identity; value added matters	Real GDP +0.2% (2025, provisional)
Tax	CBAM receipts are sale proceeds, not tax	ETS auctions, all sectors, EUR 5.4 bn (2025); CBAM sales from Feb 2027
C	Pass-through to final goods may cut spending	Consumption +1.3% real (2025)
I	Clean-plant investment up; downstream may fall	Fixed investment −0.6% real (2025)
G	Support and infrastructure; subsidies are not all G	Government consumption +1.7% real (2025)

- **Backup detail for the rows (speaker notes, A6, A12):** X also: Chapter 72 to outside the EU 7.2 to 4.7 Mt (2017–25). M also: direct 3,342 to 1,464 kt; Jan–Aug 2026 direct +7.7% (provisional). NX also: outside the EU +0.8 Mt by origin; national 2025 exports −0.9%, imports +2.9% real, net-export contribution −1.5 pp. Q also: market supply of rolled steel 40.8 to 29.1 Mt; services indirect, no series. P also: import unit value EUR 736 to 1,087/t (2019–25), mix-affected; services no series. Tax also: all to the climate fund by law; ETS receipts say nothing about steel's own payments (steel mostly gets free allocation); corporate and payroll effects not measurable. G also: aid commitments are transfers; climate-fund plan 2027 EUR 38.9 bn. I also: announced and paid-out aid are not an investment effect (S9).
- **Footer:** growth rates are the Eurostat / Destatis vintage of 21 Oct 2026 (after the Destatis revision of 25 Aug), provisional, whole-economy context and not CBAM effects; the January first estimate differed (government +1.5, investment −0.5, exports −0.3, imports +3.6, consumption +1.4). Does not show any estimated effect.
- **Label:** Observed data, with Context-only and Gap chips per row. **Chips lit:** all 11.

S11 · 1.0 min · Costs and conditions: what has to go right

- **Message:** Cleaner steel is still a premium or a gap in many settings, so any long-term benefit is conditional on clean power and hydrogen, finance, delivery and regional adjustment.
- **On the slide:** three cards and a banner. Price gap: IEA says hydrogen cost gaps often remain even with policy support [U20]; the IEA steel and cement report stresses buyer commitments and public policy [U19]; Salzgitter says new energy inputs currently cost more than coal [U12]. Capital and delivery: Duisburg has committed support of up to about EUR 2 bn (EUR 550 m grant plus up to EUR 1.45 bn conditional operating payment) against a company-stated plant cost of about EUR 3 bn (reported rising), and starts on natural gas; Stegra needs more capital to finish Boden, amount not published, no start date, new chief executive from 1 Nov 2026 [U16]; EWE covers about 6.5% of SALCOS hydrogen need [U13]. People and regions: Thyssenkrupp's plan of 25 Nov 2024 is about 5,000 job reductions by 2030 plus about 6,000 transferred or sold; the company cites weak competitiveness and cheap imports; it pre-dates definitive CBAM and is not attributed to the policy [U9]. Banner: we do not say green steel is cheaper.
- **Footer:** also at work: energy prices, COVID, sanctions, the separate steel quota (A5), state aid (about EUR 5.9 bn live commitments, A11), the planned industrial power price (EUR 3.8 bn, 2026–28; eligibility of primary steel not checked), and free-allocation rules (a 17 Jul 2026 Commission proposal would slow the phase-out of free allocation in CBAM sectors; not law). Backup A8, A9, A10, A12.
- **Label:** mixed: Company announcement and published analysis (IEA); no measured finding.

5. CH3 — Conclusion and suggestion (2.5 min)

S12 · 1.0 min · Conclusion: consistent with a move toward cleaner steel; causation not shown

- **Message:** The record is consistent with a move toward cleaner steel, but causation, a net German gain, cost parity, delivered

tonnes and a fall in CO2 per tonne are not shown. The assigned title, "can strengthen", is a team judgement conditional on the list below.

- **On the slide:** verdict banner: the record is consistent with a move toward cleaner steel; ETS and CBAM causation and the net German benefit are not shown. Three columns.
- CONSISTENT WITH: the carbon-cost comparison now includes embedded emissions [U3] (Policy rule); dated commitments 2021–26 [U10][U12][U13][U14][U15] (Company announcement; the flagship aid approvals pre-date reporting but not the 2021 proposal or the ETS price rise); OECD: about 10% emissions cut in 2005–12 with no measured harm to regulated firms, earlier period, sample not specified [U30] (Published research).
- NOT SHOWN: a sustained fall in CO2 per tonne (electric-arc share 29.2% in 1999, 27.7–32% in 2020–25, 30.5% in 2025; ETS-reported emissions divided by crude steel, derived, 0.84 in 2013, 0.90 in 2019, 0.88 in 2025; WV Stahl specific CO2 1.37 in 2019 and 1.39 in 2025); causal size; net German GDP or welfare gain (scenario studies: smaller than sectoral effects, assumption-sensitive [S19]); cheaper green steel; delivered tonnes; tax and net budget effects; paid-CBAM effects (first payments 2027). Counter-evidence: ArcelorMittal withdrew (approved aid unused), SALCOS slipped to 2027, Stegra needs capital.
- CONDITIONAL ON: affordable clean power and hydrogen; finance; new low-carbon plants running (first output 2027 or later); verified emissions data; workable administration [U12][U20].
- Context line, no steel inference: covered EU ETS emissions about 50% below 2005 in 2024, an EU aggregate led by the power sector [U29].
- **Say:** German ETS-reported emissions of iron and steel installations fell about 16% between 2019 and 2025 while crude steel fell about 14%, so the fall largely follows output; per tonne there is no clear change (ETS-derived −2.5%, WV Stahl +1.5%, within noise).
- **Label:** Team judgement.

S13 · 1.5 min · Suggestion: keep the framework, change the delivery

- **Message:** Keep a predictable ETS plus CBAM framework, improve how it is delivered, and reconsider only if monitoring shows leakage, losses that exceed benefits, or no sustained fall in emissions per tonne.
- **Addressed to:** EU legislators and the Commission (rules); the German government and the emissions-trading authority DEHSt (support, monitoring, retraining).
- **On the slide:** three columns. Continue verdict in one sentence: continue, with the changes below; nothing to drop now.
- KEEP: ETS + CBAM, and support the Commission's proposal of 17 Jul 2026 to slow the phase-out of free allocation in CBAM sectors (more allocation kept in every year from 2028, 15% still granted in 2034–37, ending in 2038 instead of 2034; not yet law) [S9]. Reasons: the first low-carbon plants only start in 2027 or later, and free allocation covered about 94% of the emissions of German iron-and-steel installations in 2024 (one year; 83.5–108% in 2019–24), so a faster phase-out would remove protection before clean capacity exists. Condition: the same proposal makes all free allocation from 2031 depend on a decarbonisation investment plan, and we support keeping that link. Trade-offs: carbon costs align more slowly at the border, importers pay a smaller CBAM charge for longer, auction receipts may be lower, and the past record shows no fall in CO2 per tonne under near-full free allocation (S12), so the case rests on conditions still to come. Also keep climate design apart from trade defence such as the steel quota [U6]; proportionate treatment of small importers, since the 50-tonne threshold would exempt 0.87% of embedded emissions (Commission assessment of 30 Sep 2026, limit 1%) [S23].
- CHANGE: (1) emissions data, verification and transparent defaults [S11]; (2) clean power, hydrogen and scrap infrastructure with evaluated support and workable conditions (the hydrogen requirement of the Duisburg aid was reported relaxed in Aug 2026); (3) monitor downstream users before any extension, which is a proposal only: Commission 17 Dec 2025 (about 180 products,

from 1 Jan 2028), Council position 12 Jun 2026, Parliament position 15 Sep 2026 (464–50–159), negotiations under way, not law [U32][S24]; (4) retraining and regional adjustment [U9]; (5) keep carbon receipts working for the transition: ETS receipts go to the climate fund by law, the 2027 draft budget would divert EUR 2.7 bn to the core budget (Bundestag vote planned 27 Nov 2026), and CBAM receipts have no earmark.

- DROP OR REVISE IF: monitoring shows leakage or resource shuffling; downstream and regional losses exceed benefits; no sustained fall in emissions per tonne; or the longer free-allocation period is not matched by new low-carbon capacity starting, which removes the reason for the stretch. The data for these triggers start in 2027 or later, after the Council (target December 2026) and Parliament positions, so we ask for a review clause in the law, not a precondition. The trigger is qualitative: baseline 2025 (0.88 t CO2 per tonne of crude steel, ETS-derived, direct emissions only; 1.39 t including purchased power, WV Stahl), compared at similar utilisation because 2025 utilisation was below 70%.
- Monitor strip: physical trade | input prices | domestic value added | realised investment | verified emissions per tonne | use of carbon receipts. Next data points: Eurostat update 13 Nov, Bundestag vote 27 Nov 2026, certificate sales 1 Feb 2027, surrender 30 Sep 2027.
- **Speaker note, stretch (not on the slide; model-based, order of magnitude):** the proposal raises the share of free allocation still granted (the "CBAM factor" of the Directive text) in every year from 2028: +1.5 points in 2028, rising to +13 in 2033, then +15 in 2034–37 (for example 59% instead of 51.5% in 2030). One point is worth about EUR 27 m a year for German iron and steel (31.2 Mt adjusted allocation in 2024 × EUR 85.10, the EEX auction of 20 Oct 2026); the extra 15 points in 2034–37 are worth about EUR 0.4 bn a year, and all extra allocation in 2028–37 about 33 Mt, about EUR 2.8 bn, at constant volume and price and ignoring benchmark updates. Views differ: EUROFER says the postponement does not remove investment uncertainty and misses a solution for exports under CBAM; Fertilizers Europe welcomed it; Cement Europe does not support postponing; EEB

and Bellona say the wider package weakens investment incentives, mainly through a lower annual cap reduction [EPRS briefing 791491]. The Commission's stated aim is to reduce carbon-leakage risk [Council ST 13435/26], which the deck does not evidence: CBAM covers imports only, Chapter 72 exports to outside the EU were 4.7 Mt in 2025, about three times direct imports (1.46 Mt), and the by-origin share of imports shows no visible shift (S7). If the final text differs from the proposal, our suggestion applies to the principle: a conditional, time-limited slowdown.

- **Label:** Team judgement.

6. German primary sources: what was verified

Source	Checked	Result for the deck
WV Stahl (steel federation), Daten und Fakten zur Stahlindustrie in Deutschland, 2025 and 2026 editions	Dataset A (crude steel 2017–24) and Dataset B (steel mill products trade, p. 30) number by number	Dataset A matches 8 of 8; 2025 = 34.1 Mt (later revised to 34.3 Mt). Dataset B matches 16 of 16 in the 2025 edition; the 2026 edition revises 2024 (exports 23.3, imports 19.0 Mt) and adds 2025 (22.4 / 20.5 Mt, balance +1.9 Mt). The workbook's derived balance differs from WV Stahl's printed one by 0.1 Mt in five years, so the printed one is used. New: market supply of rolled steel 2012–2025; EU-27 trade with third countries; indirect steel trade; direct steel employment about 80,000.

Source	Checked	Result for the deck
Destatis (GENESIS)	Chapter 72 trade by EU / non-EU, production index, employment, producer prices, national accounts	Eurostat's manufacturing table is Destatis data (31 of 31 years match). Non-EU exports equal Comext within 0.1% in 2017–25 (up to 1.5% in 2008–16). By-origin non-EU imports are far above direct imports (3.9 versus 1.5 Mt, 2025). Producer price index pig iron and steel (2021 = 100): 108.0 (2024), 101.2 (2025), 107.9 (Sep 2026). 2025 national accounts revised on 25 Aug 2026. The "5.5 million workers" in the chat is the 50+ employee establishment universe (7.4 million is the national-accounts figure).
Bundesnetzagentur SMARD	Industrial electricity price series behind slide 5	Five printed numbers match; the 2020 value was from the other series (see section 1). SMARD is a model; the series was re-modelled since February 2025 (2024: 16.77 became 16.12), so slides name the vintage.

Source	Checked	Result for the deck
German government and company reports on aid	tkH2Steel, SALCOS, carbon contracts, industrial power price	tkH2Steel: EUR 550 m grant plus up to EUR 1.45 bn conditional payment (EU decision 20 Jul 2023), about EUR 2 bn committed by Bund and NRW. SALCOS: EUR 1 bn (EU decision 4 Oct 2022), about EUR 1.3 bn since Feb 2026, EUR 1.076 bn paid by 30 Jun 2026. Carbon contracts: EUR 5 bn, 15 years, at least 50% reduction within four years and 85% at the end (EU decision 7 May 2026); round 2 awards expected Nov–Dec 2026. Industrial power price EUR 3.8 bn (2026–28). Neither flagship holds a carbon contract.

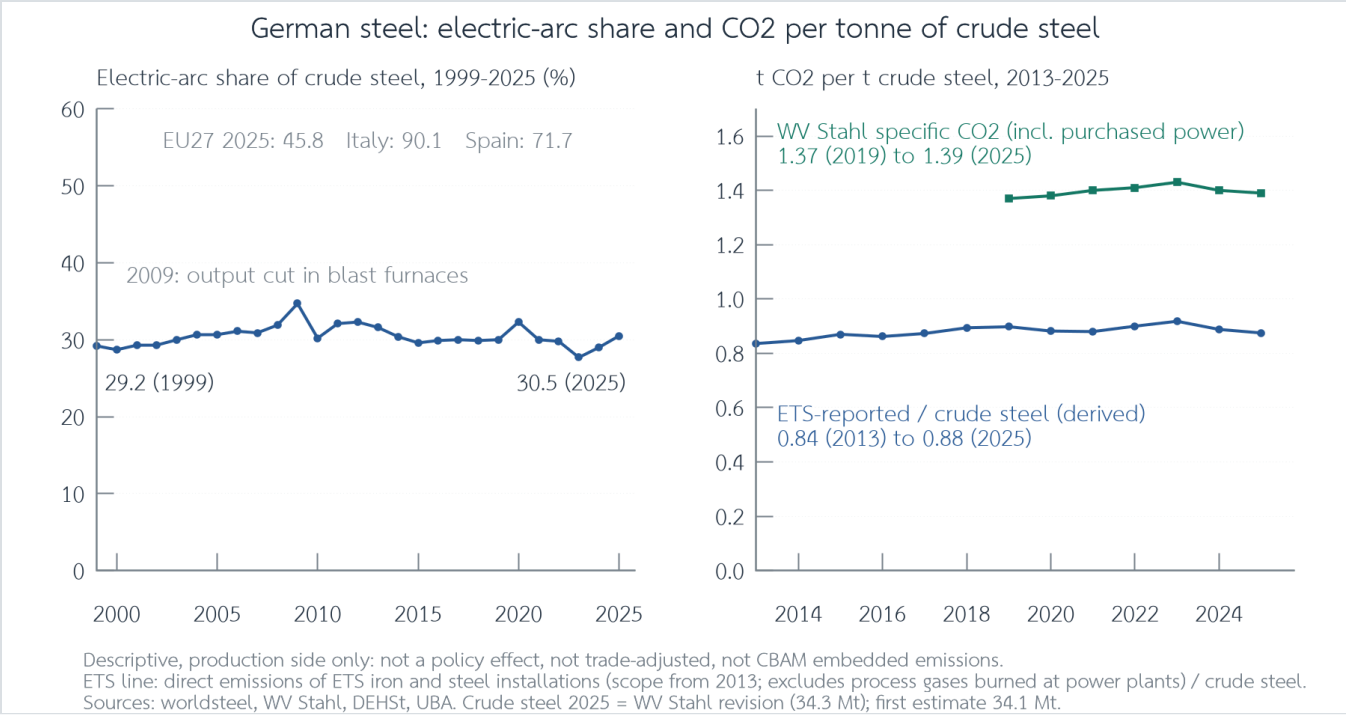
Cautions from this pass: the aid payout per project is classified by the federal government, so "paid" comes from company reports; the August 2026 relaxation of the Duisburg hydrogen conditions is known only from trade press relaying Reuters; BMWE pages and EUR-Lex were blocked for automated access and were not bypassed; the by-origin import series is a team construction from Destatis tables (all partners minus EU27).

7. Backup pages (not timed)

#	Page	Answers
A1	Fast answers on one page	Why start before 2005? Why no DiD? Did CBAM force the contracts? Is DRI always fossil-free? Is CBAM a fixed tariff? Is CBAM revenue German tax? (No: certificate sale proceeds kept by the selling Member State, first from Feb 2027; 2027 draft budget EUR 0.38 bn.)

#	Page	Answers
A2	Claims we corrected	14% share, 5.5 m / 20.8% (the 50+ employee universe), 47% quota, 11,000 jobs, "GDP barely changed", cheaper green steel, contracts = delivery, **electricity "nearly triple"**, **"full hydrogen by 2029"**, **SALCOS EUR 1 bn**
A3	Worked example, phase-in table, certificate prices	Quarter, publication date, price (Q4 due 4 Jan 2027); current-law factors beside the 17 Jul 2026 proposal; latest EUA auction EUR 85.10 on 20 Oct 2026 (latest in the EEX file) as context only
A4	Coverage, exempt origins, 50-tonne threshold, who pays	The Member State where the declarant is established collects the receipts
A5	The steel quota, sanctions and dumping are not CBAM	Reg. (EU) 2026/1384: 18,345,922 t, 50% duty, about 47% below the 2024 base
A6	NX, BOP and the tax row	Full WV Stahl series 2017–25 with printed balance and market supply of rolled steel; extra-EU balance tables (direct 1999–2025, by origin 2008–2025); Jan–Aug 2026 net balances by concept (direct −27%, by origin Jan–Jul far lower, all-partner Chapter 72 +16%); WV Stahl versus Chapter 72 scope table; concept table (customs goods exports EUR 1,564 bn, BOP goods surplus EUR 184 bn, current account about EUR 203 bn, national-accounts net exports EUR 156.7 bn to 100.3 bn); ETS auction receipts 2012–2025; CBAM revenue rules; tax channels not measurable
A7	Reading the import series	2005 rise (semi-finished products), 2022 unit values, DRI (7203) direct versus by origin, Jan–Aug 2026 table (Comext and Destatis), materiality shares, origin share flat at 16–18%

#	Page	Answers
A8	What still enters from outside the EU; has German steel become cleaner?	2025 import composition; the chart below; route emissions per tonne
A9	Is green steel cheaper?	Premium range, scrap, aluminium, composites (BaseNotes-2, provisional)
A10	Domestic production-cost data	Electricity (two SMARD series, invoiced cross-checks from Destatis and Eurostat); ETS price path; free allocation about 94%; hydrogen, capex and opex gap remain provisional (BaseNotes-3)
A11	Green support criteria and exemptions	Verified table of aid by project (amount, decision date, split, paid); carbon contracts; industrial power price; imported steel is generally not a target of these schemes (workbook reading, not re-verified); no CBAM green-steel exemption; EUR 12.5 bn / 11.5 Mt unverified
A12	Jobs, regions, skills, wider sectors	Manufacturing employment 7,262 thousand (2025, provisional); establishments with 50+ employees 5.43 million; basic metals 245 thousand (2024) against 268 thousand (2015); WV Stahl direct steel employment about 80,000 (a narrower group); basic metals are about 0.5% of GVA (2024)
A13	Company-case fine print	Did CBAM force the contracts? Stegra status as of 22 Oct 2026; Volvo–SSAB and Outokumpu; retire "full hydrogen by 2029"; granted versus paid; a Gephi map with all eleven nodes is exported (`graphs/out/a13_company_network_full.png`) for use here if the table is cut
A14	Spare theory diagrams	Fig. 3 subsidy, Fig. 4 demand shift (relabel "one fixed point in time"), Fig. 5 offshoring, Fig. 6 labour, original Fig. 2
A15	Source list	Section 9 below



8. What is still open, and risks

1. **Lecturer.** Send the retitle and the 15-minute plan through the channel used for the topic (4–7 Oct) and record the answer; the check of the hand-drawn theory graphs needs the on-site class. Each diagram slide has a text fallback with the same minutes.
2. **Facts that may move before the talks (no refresh is scheduled).** Eurostat releases September data and revisions on 13 Nov (Jan–Aug 2026 figures will restate and the headline moves with the cutoff: +6.1% at July, +7.7% at August); WV Stahl September crude steel; SMARD and Destatis monthly updates; the Bundestag budget committee meets 12 Nov (vote 27 Nov); round-2 carbon-contract awards (Nov–Dec); re-check Stegra after the new chief executive starts on 1 Nov, the downstream-extension talks (the 20 Oct trilogue outcome was not found) and the ETS review (Council target December). The Q4 certificate price will not exist until 4 Jan 2027.
3. **Freeze date.** The chat's 14/11 final confirm is 2–4 days before the 16–18/11 slots and Member D leaves Thailand in early November; suggested content freeze about 1 Nov; the final confirm is 14 Nov.
4. **Team decisions:** recorded on 22 Oct 2026 from the requester: support slowing the phase-out of free allocation in CBAM sectors

as proposed on 17 Jul 2026 (S13 KEEP). The LINE chat holds no discussion of it, so confirm it with all members before presenting. Still open: whether the "reconsider" triggers get thresholds and dates; the Council (December) and Parliament (14 December plenary forecast) stages may change the proposal, and S13 then applies to the principle (a conditional, time-limited slowdown).

5. **Lecture PDFs** are not archived locally; re-check page references (G1 pp. 13, 15–17; G2 pp. 5–7; G3 pp. 2, 4–5, 10). **Fig. 4 ruling** (partial shift in the backup) to confirm with Member D.
6. **Not verified:** EUR 12.5 bn total project cost and 11.5 Mt a year; the August 2026 relaxation of the Duisburg hydrogen conditions (press only); SMARD's consumption band; hydrogen, ETS-price and capex rows in BaseNotes-3; any partner-country (China, Russia, Turkey, India) split of imports, which the repository does not hold.

Main risks: theory graphs rejected (fallbacks keep the minutes); slide 8 misread as national-accounts NX or competitiveness (neutral wording, import concept named every time); causation creep (use "consistent with"); stale facts by 16–18 Nov; dense slides S3, S7–S10 (cut order if long: S3 purpose row, the S7 unit-value line, on S10 read only the trade block and the tax row; the S7 and S8 detail lives in speaker notes); company wording (use the S9 phrasing verbatim). Questions the deck now answers with data: evidence on cleaner steel (no clear per-tonne change), tax (ETS receipts observed but not steel-specific, CBAM from Feb 2027, corporate and payroll not measurable), materiality (8% of German Chapter 72 import tonnes direct, 17% by origin), exports (pulled and cross-checked against Destatis), domestic prices and demand.

20-minute stretch plan if the course brief's 20 minutes is enforced: add 5.0 minutes: S3 +0.5, S5 +0.5, S7 +0.5, S8 +0.5, S9 +1.0 (Volvo–SSAB, Outokumpu, BMW detail), S10 +0.5, S11 +1.0, S13 +0.5.

Suggested speaker split (not recorded in the chat): CH1 (S1–S4); theory, manufacturing and imports (S5–S7); NX, firms, checklist and costs (S8–S11); CH3 (S12–S13) plus the Q&A lead using A1. Rehearse to the checkpoints 4.0 / 12.5 / 15.0 minutes.

9. A15 — Source list

Tags used on the slides.

Tag	Source
Dataset A	WV Stahl, crude steel production 2017–2025 (Daten und Fakten, 2026 edition, p. 9; revision tables)
Dataset B	WV Stahl, steel mill products exports and imports 2017–2025 (Daten und Fakten, p. 30)
Dataset C	Eurostat Comext DS-045409, HS Chapter 72 imports from outside the EU, with unit values
Dataset D	Comext HS 7208 imports, 2019–2025
Dataset E	European Commission CBAM certificate prices 2026 (Q1–Q3)
Dataset F	Eurostat / Destatis national-accounts growth 2025, vintage of 21 Oct 2026
BaseNotes-1 to -5	Detailed English notes in the base workbook (subsidies; alternative supply and prices; domestic costs; trade-data method; jobs and GDP); no source IDs, treated as provisional
Comext, Destatis, WV Stahl, SMARD	Tables and READMEs under `data/` in the repository, with query parameters and retrieval dates

Workbook source tags.

Tag	Source
G1	Assoc. Prof. Attasuda Lerskullawat — Reason for Domestic Protection, supplied lecture PDF: page 13 (Production/Consumption Effect), pages 15–17 (externalities).
G2	Assoc. Prof. Attasuda Lerskullawat — Export Policy, supplied lecture PDF: pages 5–7, Production Subsidy.
G3	Assoc. Prof. Attasuda Lerskullawat — Offshoring of Goods and Services, supplied lecture PDF: pages 2, 4–5 and 10. Page references count PDF pages, including the title page.
G4	European Commission — About the EU Emissions Trading System
G5	European Commission — Carbon Border Adjustment Mechanism
S1	European Commission — CBAM overview
S2	Council of the EU — CBAM policy explanation
S3	European Commission — Original CBAM proposal, 14 July 2021
S4	Regulation (EU) 2023/956 — consolidated CBAM law, including Annexes I–III
S5	Regulation (EU) 2025/2083 — adopted simplification amendment
S6	European Commission — CBAM definitive regime
S7	German Environment Agency — CBAM simplification and German implementation

Tag	Source
S8	European Commission — Official CBAM certificate prices
S9	Directive (EU) 2023/959 — EU ETS amendment and CBAM factor schedule
S10	Implementing Regulation (EU) 2025/2620 — free-allocation adjustment
S11	European Commission — Current CBAM legislation, benchmarks and steel guidance
S12	German Steel Federation — Facts and Figures on the Steel Industry 2025 (German original; pp. 9, 30)
S13	World Steel Association — December 2025 and annual 2025 steel production
S14	Eurostat — International trade in goods database / Comext
S15	Eurostat — Trade data definitions and methodology
S16	Destatis — GDP up 0.2% in 2025, first annual estimate (15 January 2026)
S17	Destatis — Full-year 2025 merchandise trade (6 February 2026)
S18	Deutsche Bundesbank — German balance of payments in 2025, Monthly Report March 2026
S19	German Environment Agency — Models for EU ETS and CBAM, Summary, Climate Change 77/2025
S20	WTO — Environmental exceptions under GATT Article XX
S21	European Commission — CBAM application review, COM(2025) 783, annexes (Annex 4, Figure 2)
S22	World Steel Association — World Steel in Figures 2026
S23	European Commission — CBAM communication and news, including September 2026 threshold assessment
S24	European Commission — Consultation on downstream extension (2 July 2025)
U1	European Commission — EU ETS: operation, history and phases
U2	European Commission — Carbon leakage and free allocation
U3	European Commission — CBAM overview
U4	Consolidated CBAM Regulation (EU) 2023/956, including current product scope
U5	European Commission — Definitive CBAM regime
U6	Council of the EU — Adoption of the new steel trade measure, 8 June 2026
U7	European Commission — Steel quota factsheet: 18.3 million tonnes
U8	Council of the EU — Steel measure agreement and 47% comparison, 13 April 2026
U9	Thyssenkrupp Steel — Restructuring plan, 25 November 2024
U10	Volkswagen Group — Thyssenkrupp low-carbon steel MoU, 22 October 2024
U11	Thyssenkrupp Steel — bluemint product family and recycling route
U12	Salzgitter — SALCOS technology, current milestones and cost conditions
U13	Salzgitter — EWE green hydrogen supply contract, 9 June 2026
U14	Mercedes-Benz — 2023 company history, including the 50,000-tonne supply deal
U15	BMW Group — H2 Green Steel agreement, 20 October 2021
U16	Stegra — Project review and further capital requirement, 30 September 2026
U17	Volvo Group — Vehicle using SSAB fossil-free steel, 13 October 2021
U18	Outokumpu — 2025 environmental and recycled-content figures
U19	IEA — Demand and supply measures for the steel and cement transition (2025), executive summary
U20	IEA — Global Hydrogen Review 2026: cost acceptability
U21	IEA — Steel production routes and transition, Breakthrough Agenda Report 2025

Tag	Source
U22	Eurostat — Manufacturing GVA, nama_10_a64
U23	Eurostat — Annual industrial production, sts_inpr_a
U24	Eurostat — International trade in goods / Comext database
U25	Eurostat — Detailed trade definitions and methodology
U26	Eurostat — Sold industrial production, including Germany's 26% share in 2025
U27	Eurostat — Manufacturing businesses: 2023 EU employment and business-economy scope
U28	Eurostat — Employment by industry, nama_10_a64_e
U29	European Commission — 2025 Carbon Market Report: emissions from power and industry
U30	OECD — ETS emissions and firm economic performance (2018 working paper)
U31	European Commission — Shipping sector emissions and EU ETS
U32	European Parliament — Downstream CBAM proposal status, 15 September 2026
U33	Council of the EU — Russia sanctions and import restrictions
U34	European Commission — CBAM steel guidance and implementing legislation
U35	European Commission — Certificate prices and February 2027 sales

Sources added on 22 October 2026 (raw pages and tables in the repository under data/): Eurostat Comext DS-045409 (annual and monthly, vintage 16 Oct 2026); Eurostat nama_10_a64, nama_10_a64_e, sts_inpr_a, nama_10_gdp, gov_10a_taxag, env_ac_tax; Destatis GENESIS tables 51000-0005/-0006/-0007, 42153-0001/-0002, 42111-0001/-0003, 61241-0001 to -0004, 81000-0019/-0029/-0123 and the national-accounts revision of 25 Aug 2026; WV Stahl Daten und Fakten zur Stahlindustrie in Deutschland (2025 and 2026 editions), annual crude-steel revision tables 2021–2025, scrap report of 14 Oct 2026; worldsteel statistical yearbooks and World Steel in Figures 2026; DEHSt / UBA emissions and auction data; EEX auction reports; Bundesnetzagentur SMARD (article of 11 Feb 2025, data file of 22 Oct 2026) and Monitoringbericht 2025; Destatis 61243-0005 and Eurostat nrg_pc_205; BDEW Strompreisanalyse 08/2026; European Commission CBAM price page and legislation pages; Regulation (EU) 2026/1384; Regulation (EU) 2025/2083; Directive (EU) 2023/959; COM(2025) 989; COM(2026) 616; Council documents 10423/26, 13435/26, 13922/26; European Commission aid decisions IP/22/5968, IP/23/3928, IP/26/997; Bundestag Drucksachen 20/12166, 21/3563, 21/7300, 21/7837, 21/7860; Salzgitter H1 2026 report; thyssenkrupp interim report 9M 2025/26; Stegra releases of 30 Sep and 6 Oct 2026; BMW press release of 20 Oct 2021.